

Basic Understanding on DevSecOps

With

AWS

What's in this series?

- Real-world usage of DevOps & DevSecOps tools.
- How tools like Jenkins, Docker, Kubernetes, and Terraform work together.
- Security practices using tools like SonarQube and Trivy.
- Simple examples to help beginners understand.
- A full month of daily learning posts!

Let's Start with Day-1

What is DevSecOps?

- DevSecOps stands for Development, Security, and Operations.
- It's a culture and approach where security is added into every stage of software development, rather than waiting till the end.

Why it matters:

Traditionally, security checks used to happen only after development. But by then, fixing issues was costly and time-consuming. DevSecOps helps detect security issues early, saving time, money, and reducing risk.

Example:

A team using DevSecOps runs security scans (like Trivy or SonarQube) every time code is pushed to GitHub. If there's a security flaw, the build fails — and developers fix it before the code goes further.

What is SDLC (Software Development Life Cycle)?

SDLC is the **step-by-step process** followed to develop software:

- Planning
- Design
- Development
- Testing
- Deployment
- Maintenance

Why it matters:

Having a clear SDLC helps teams build software in a structured and organized way. It reduces errors, improves quality, and keeps everyone on the same page.

Example:

Imagine building a mobile app. First, you gather requirements (planning), then design the screens (design), write code (development), test the app (testing), release it on the app store (deployment), and then keep fixing issues and updating features (maintenance).

What is CI/CD?

CI/CD stands for:

- **Continuous Integration (CI):** Developers merge their code frequently, and the system automatically builds and tests the app.
- **Continuous Delivery/Deployment (CD):** After testing, the app is **automatically deployed** to staging or production.

Why it matters:

CI/CD ensures faster and more reliable releases. You can catch bugs early and release updates more often with confidence.

Example:

Using Jenkins or GitHub Actions, code changes can be automatically built, tested, and deployed to an AWS server every time a developer makes a change.

Here are the tools and topics I'll be covering, along with how they are used in a DevSecOps pipeline:

Linux

Used for managing servers, running containers, scripting, and almost everything in DevOps. Knowing basic commands is crucial for DevOps engineers.

GitHub

Version control tool used to store, manage, and collaborate on source code. Integrates with CI/CD tools to trigger builds when code is pushed.

AWS (Amazon Web Services)

Cloud platform to deploy, scale, and manage applications. Commonly used services include VPC, EC2, S3, IAM etc., in DevOps workflows.

SonarQube – Code Quality & Security Testing

Analyzes source code to detect bugs, code smells, and vulnerabilities. Ensures code quality before deployment.

Build Tools – Maven & NPM

Used to build projects and manage dependencies:

- **Maven:** Mostly for Java applications
 - **NPM:** For Node.js applications
-

Docker – Containerization

Packages applications into lightweight containers. Helps run apps consistently across environments.

Trivy – Docker Image Scanning

Scans Docker images for vulnerabilities. Helps in identifying risks before pushing images to production.

Kubernetes – Container Orchestration

Manages and scales Docker containers. Automates deployment, scaling, and management of containerized apps.

Ansible – Configuration Management

Used to automate server setup, deployments, and updates. Ensures consistent environments across multiple servers.

Jenkins – CI/CD Automation

Automates code builds, testing, and deployments. Acts as the backbone for CI/CD pipelines in most DevOps projects.

Terraform – Infrastructure as Code

Used to provision and manage cloud infrastructure using code. Helps automate and scale cloud resources (like AWS, Azure, GCP).

Thank You